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KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF EDUCATION AND
BACHELOR OF ARTS
AGE 201/211: PHYSICAL GEOGRAPHY II

1ST SEMESTER 2021/2022

TIME: 2 HOURS

INSTRUCTIONS:

Answer **question ONE** and any **other two** question, i.e one question from **section B** and another one from **section C**

Section A: Compulsory (30 marks)

Question 1

- a. Differentiate between Coalescence model and Bergeron-Findeise process (3 marks)
- b. Suggest three ways in which flood frequency analyses may be useful (3 marks)
- c. (i) What is a river gauging station (RGS)?
(ii) Which is the best location for RGS? (3 marks)
- d. Differentiate between connate and juvenile water (3 marks)
- e. Calculate the porosity of a 250 g rock that contains 65 g of water when 55% of the pores are full of water (3 marks)
- f. Explain the types of soil moisture in relation to their availability for use by plants (3marks)
- g. Highlight three theories on why Rift valley lakes are rising in sea levels (3 marks)
- h. Explain how glacier is formed (3 marks)
- i. Highlight three hazards associated with slope processes (3 marks)
- j. State three factors that make wind an effective agent of desert erosion (3 marks)

SECTION B: HYDROSPHERE (choose one question) (20 marks)

Question 2

- a. A soil sample has a weight of 0.7 kg and the volume was found to be $3.5 \times 10^{-4} \text{ m}^3$. After drying out the weight was reduced to 0.6 kg. The particle specific gravity test gave 2.6.

Determine the following:

- (i) Moisture content

- (ii) Dry density
- (iii) Void ration
- (iv) Bulk density
- (v) Porosity
- (vi) Degree of saturation

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(12 marks)

- b. State **FOUR** challenges of rainwater harvesting in your home county (8 marks)

Question 3

- a. In its effort to rehabilitate the Mau forest complex, the government of Kenya initiated a policy of evicting forest settlers. A politician was heard telling some of his Mau residents not to vacate the forest since *'trees do not bring water..... rains come from God'*. Critically analyze the politician's statement.

(12 marks)

- b. Discuss FOUR ways in which ground water is polluted in your home county(8 marks)

SECTION C: LITHOSPHERE (choose one question) (20marks)

Question 4

- a. Explain FIVE factors that can interrupt Morris Davis cycle of erosion (10 marks)
- b. Using diagrams, describe FIVE features formed by wave erosion at the Indian Ocean

(10 marks)

Question 5

- a. Describe FIVE factors affecting slope stability (10 marks)
- b. Using FIVE examples, explain how weathering has affected the economic activities in the county of your choice. (10 marks)

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